

NBA Play-by-Play Data Dictionary

cdnnba + shotdetail (2020-2025, regular season + playoffs)

Rows: 4,068,306 | Columns: 66 | Source: cdn.nba.com + stats.nba.com shot zones

Column Summary

#	Column	Type	Nulls	Uniq	Description
1	actionNumber	Int64	0%	951	Sequence number of the event within the game
2	clock	String	0%	1,261	Time remaining in the period (ISO 8601 duration, e.g...
3	timeActual	String	0%3,095,216		Wall-clock timestamp of the event (UTC)
4	period	Int64	0%	7	Period number (1-4 = quarters, 5+ = overtime)
5	periodType	String	0%	2	Whether the period is regulation or overtime
6	actionType	String	0%	19	Primary event category (shot, foul, rebound, etc.)
7	subType	String	2%	64	Secondary event classification (shot type, foul type...
8	qualifiers	String	0%	84	Comma-separated event tags (fastbreak, 2ndchance, po...
9	personId	Int64	0%	1,158	NBA player ID of the primary actor (0 for team/game ...
10	x	Float64	68%	5,221	Court x-coordinate of the action (percentage, 0-100)
11	y	Float64	68%	7,605	Court y-coordinate of the action (percentage, 0-100)
12	possession	Int64	0%	31	Team ID of the team with possession (0 for dead ball)
13	scoreHome	Int64	0%	167	Home team score at time of event
14	scoreAway	Int64	0%	170	Away team score at time of event
15	edited	String	0%2,662,311		Timestamp of last edit to this event record
16	orderNumber	Int64	0%	11,984	Global ordering key for events within a game
17	xLegacy	Int64	68%	502	Legacy court x-coordinate (in tenths of feet from ce...
18	yLegacy	Int64	68%	855	Legacy court y-coordinate (in tenths of feet from ba...
19	isFieldGoal	Int64	0%	2	1 if event is a field goal attempt, 0 otherwise
20	side	String	68%	3	Which side of the court the action occurred on
21	description	String	1%	910,094	Human-readable text description of the event
22	personIdsFilter	String	0%	215,851	Pipe-separated list of all player IDs involved in th...
23	teamId	Int64	2%	31	NBA team ID of the primary actor
24	teamTricode	String	2%	31	Three-letter team abbreviation (e.g. BOS, LAL)
25	descriptor	String	73%	67	Shot style modifier (pullup, driving, step back, fad...
26	jumpBallRecoveredName	String	100%	741	Name of player who recovered the jump ball tip
27	jumpBallWonPlayerName	String	100%	560	Name of player who won the jump ball
28	jumpBallWonPersonId	Int64	100%	670	Player ID of jump ball winner
29	jumpBallLostPlayerName	String	100%	613	Name of player who lost the jump ball
30	jumpBallLostPersonId	Int64	100%	728	Player ID of jump ball loser
31	officialId	Int64	90%	105	ID of the referee involved (fouls, violations, reviews)
32	turnoverTotal	Int64	95%	13	Player's running turnover total for the game
33	playerName	String	8%	868	Last name of the primary player
34	playerNameI	String	8%	1,048	Abbreviated name of the primary player (e.g. J. Brown)
35	foulPersonalTotal	Int64	93%	8	Player's running personal foul total for the game
36	foulTechnicalTotal	Int64	93%	5	Player's running technical foul total for the game
37	foulDrawnPlayerName	String	93%	820	Name of the player who drew the foul
38	foulDrawnPersonId	Int64	93%	1,014	Player ID of the player who drew the foul
39	shotResult	String	60%	3	Whether the shot was made or missed (field goals only)
40	pointsTotal	Int64	79%	74	Player's running points total for the game
41	shotDistance	Float64	68%	6,119	Distance of the shot from the basket (feet)
42	assistPlayerNameInitial	String	91%	994	Abbreviated name of the assisting player
43	assistPersonId	Int64	91%	1,023	Player ID of the assisting player
44	assistTotal	Int64	91%	25	Assisting player's running assist total for the game
45	shotActionNumber	Int64	81%	889	Links a rebound/block back to the originating shot e...
46	reboundTotal	Int64	84%	32	Player's running total rebound count for the game
47	reboundDefensiveTotal	Int64	84%	25	Player's running defensive rebound total for the game
48	reboundOffensiveTotal	Int64	84%	17	Player's running offensive rebound total for the game
49	blockPlayerName	String	98%	774	Name of the player who blocked the shot
50	blockPersonId	Int64	98%	941	Player ID of the shot blocker
51	jumpBallRecoverdPersonId	Int64	100%	725	Player ID of jump ball recoverer
52	stealPlayerName	String	97%	798	Name of the player who made the steal
53	stealPersonId	Int64	97%	986	Player ID of the stealer
54	gameId	Int64	0%	7,281	Unique game identifier (e.g. 22400001)
55	value	String	100%	872	Supplemental info for instant replay / review events
56	season	Int32	0%	6	Season start year (e.g. 2024 = 2024-25 season)
57	season_type	String	0%	2	Part of season: rg = regular season, po = playoffs
58	area	String	57%	7	Shot zone area from cdn.nba.com (similar to SHOT_ZONE...
59	areaDetail	String	57%	15	Detailed shot zone with distance bucket and direction
60	isTargetScoreLastPeriod	Boolean	36%	2	Whether this event relates to a target score in the ...
61	shortFormattedClock	String	99%	721	Period clock in MM:SS format (only on some events)
62	jerseyNumber	String	99%	55	Player's jersey number (only on some events)
63	SHOT_ZONE_BASIC	String	68%	8	Shot zone classification from stats.nba.com (shotdet...
64	SHOT_ZONE_AREA	String	68%	7	Court area of the shot from stats.nba.com (shotdetai...
65	SHOT_ZONE_RANGE	String	68%	6	Distance range bucket from stats.nba.com (shotdetail...
66	shot_action_type	String	68%	49	Granular shot type from stats.nba.com (e.g. Driving ...

Column Details

actionNumber

Int64 | 0.0% null | 951 unique
Sequence number of the event within the game

clock

String | 0.0% null | 1,261 unique
Time remaining in the period (ISO 8601 duration, e.g. PT11M43.00S)

timeActual

String | 0.0% null | 3,095,216 unique
Wall-clock timestamp of the event (UTC)

period

Int64 | 0.0% null | 7 unique
Period number (1-4 = quarters, 5+ = overtime)

4	1,056,864	(26.0%)
2	1,044,570	(25.7%)
3	994,993	(24.5%)
1	945,792	(23.2%)
5	23,229	(0.6%)
6	2,732	(0.1%)
7	126	(0.0%)

periodType

String | 0.0% null | 2 unique
Whether the period is regulation or overtime

REGULAR	4,042,219	(99.4%)
OVERTIME	26,087	(0.6%)

actionType

String | 0.0% null | 19 unique
Primary event category (shot, foul, rebound, etc.)

substitution	814,662	(20.0%)
2pt	771,757	(19.0%)
rebound	757,204	(18.6%)
3pt	516,915	(12.7%)
freethrow	324,753	(8.0%)
foul	291,129	(7.2%)
turnover	203,090	(5.0%)
steal	112,052	(2.8%)
timeout	80,286	(2.0%)
block	70,559	(1.7%)
period	59,076	(1.5%)
stoppage	28,170	(0.7%)
jumpball	12,897	(0.3%)
violation	12,558	(0.3%)
game	7,282	(0.2%)
instantreplay	4,341	(0.1%)
heave	765	(0.0%)
ejection	472	(0.0%)
memo	338	(0.0%)

subType

String | 1.5% null | 64 unique

Secondary event classification (shot type, foul type, rebound type, etc.)

Jump Shot	832,851	(20.8%)
defensive	517,345	(12.9%)
in	407,331	(10.2%)
out	407,331	(10.2%)
Layup	343,754	(8.6%)
offensive	263,651	(6.6%)
personal	258,589	(6.5%)
1 of 2	133,320	(3.3%)
2 of 2	133,242	(3.3%)
	122,373	(3.1%)
full	78,779	(2.0%)
DUNK	75,327	(1.9%)
bad pass	71,585	(1.8%)
out-of-bounds	65,619	(1.6%)
1 of 1	43,679	(1.1%)
lost ball	41,960	(1.0%)
end	36,820	(0.9%)
Hook	36,737	(0.9%)
start	29,538	(0.7%)
offensive foul	23,886	(0.6%)
... and 43 more values		

qualifiers

String | 0.0% null | 84 unique

Comma-separated event tags (fastbreak, 2ndchance, pointsinthepaint, etc.)

	2,658,791	(65.4%)
pointsinthepaint	392,840	(9.7%)
team	120,387	(3.0%)
startperiod	118,188	(2.9%)
fromturnover	90,419	(2.2%)
2ndchance	82,650	(2.0%)
2freethrow	82,012	(2.0%)
pointsinthepaint, 2ndchance	66,602	(1.6%)
fastbreak	61,129	(1.5%)
team, mandatory	55,540	(1.4%)
inpenalty, 2freethrow	51,224	(1.3%)
pointsinthepaint, fromturnover	50,944	(1.3%)
1freethrow	34,143	(0.8%)
pointsinthepaint, fastbreak	29,053	(0.7%)
deadball, team	28,712	(0.7%)
pointsinthepaint, fromturnover, fastbr	22,405	(0.6%)
fastbreak, pointsinthepaint	17,430	(0.4%)
fromturnover, fastbreak	16,468	(0.4%)
2ndchance, pointsinthepaint	14,063	(0.3%)
fromturnover, 2ndchance	12,849	(0.3%)
... and 64 more values		

personId

Int64 | 0.0% null | 1,158 unique

NBA player ID of the primary actor (0 for team/game events)

x

Float64 | 68.3% null | 5,221 unique

Court x-coordinate of the action (percentage, 0-100)

y

Float64 | 68.3% null | 7,605 unique

Court y-coordinate of the action (percentage, 0-100)

possession

Int64 | 0.0% null | 31 unique

Team ID of the team with possession (0 for dead ball)

scoreHome

Int64 | 0.0% null | 167 unique
Home team score at time of event

scoreAway

Int64 | 0.0% null | 170 unique
Away team score at time of event

edited

String | 0.0% null | 2,662,311 unique
Timestamp of last edit to this event record

orderNumber

Int64 | 0.0% null | 11,984 unique
Global ordering key for events within a game

xLegacy

Int64 | 68.3% null | 502 unique
Legacy court x-coordinate (in tenths of feet from center, -250 to 250)

yLegacy

Int64 | 68.3% null | 855 unique
Legacy court y-coordinate (in tenths of feet from baseline, -50 to ~890)

isFieldGoal

Int64 | 0.0% null | 2 unique
1 if event is a field goal attempt, 0 otherwise

0	2,779,634	(68.3%)
1	1,288,672	(31.7%)

side

String | 68.3% null | 3 unique
Which side of the court the action occurred on

right	644,569	(50.0%)
left	644,103	(50.0%)

description

String | 0.8% null | 910,094 unique
Human-readable text description of the event

personIdsFilter

String | 0.0% null | 215,851 unique
Pipe-separated list of all player IDs involved in the event

teamId

Int64 | 2.4% null | 31 unique
NBA team ID of the primary actor

teamTricode

String | 2.4% null | 31 unique
Three-letter team abbreviation (e.g. BOS, LAL)

descriptor

String | 73.1% null | 67 unique

Shot style modifier (pullup, driving, step back, fadeaway, etc.)

pullup	152,160	(13.9%)
shooting	143,992	(13.1%)
driving	134,274	(12.2%)
running	81,335	(7.4%)
driving floating	66,701	(6.1%)
step back	65,133	(5.9%)
cutting	50,326	(4.6%)
driving finger roll	32,870	(3.0%)
tip	31,986	(2.9%)
turnaround	31,094	(2.8%)
floating	25,124	(2.3%)
putback	23,956	(2.2%)
fadeaway	23,733	(2.2%)
bad pass	23,455	(2.1%)
turnaround fadeaway	19,309	(1.8%)
loose ball	17,116	(1.6%)
driving floating bank	15,898	(1.5%)
running pullup	15,144	(1.4%)
alley-oop	14,929	(1.4%)
driving reverse	13,092	(1.2%)
... and 46 more values		

jumpBallRecoveredName

String | 99.7% null | 741 unique

Name of player who recovered the jump ball tip

jumpBallWonPlayerName

String | 99.7% null | 560 unique

Name of player who won the jump ball

jumpBallWonPersonId

Int64 | 99.7% null | 670 unique

Player ID of jump ball winner

jumpBallLostPlayerName

String | 99.7% null | 613 unique

Name of player who lost the jump ball

jumpBallLostPersonId

Int64 | 99.7% null | 728 unique

Player ID of jump ball loser

officialId

Int64 | 90.3% null | 105 unique

ID of the referee involved (fouls, violations, reviews)

turnoverTotal

Int64 | 95.3% null | 13 unique

Player's running turnover total for the game

playerName

String | 7.7% null | 868 unique

Last name of the primary player

playerNameI

String | 7.7% null | 1,048 unique

Abbreviated name of the primary player (e.g. J. Brown)

foulPersonalTotal

Int64 | 92.9% null | 8 unique

Player's running personal foul total for the game

foulTechnicalTotal

Int64 | 92.9% null | 5 unique
Player's running technical foul total for the game

foulDrawnPlayerName

String | 93.1% null | 820 unique
Name of the player who drew the foul

foulDrawnPersonId

Int64 | 93.1% null | 1,014 unique
Player ID of the player who drew the foul

shotResult

String | 60.3% null | 3 unique
Whether the shot was made or missed (field goals only)

Made	856,986	(53.1%)
Missed	756,439	(46.9%)

pointsTotal

Int64 | 78.9% null | 74 unique
Player's running points total for the game

shotDistance

Float64 | 68.3% null | 6,119 unique
Distance of the shot from the basket (feet)

assistPlayerNameInitial

String | 90.8% null | 994 unique
Abbreviated name of the assisting player

assistPersonId

Int64 | 90.8% null | 1,023 unique
Player ID of the assisting player

assistTotal

Int64 | 90.8% null | 25 unique
Assisting player's running assist total for the game

shotActionNumber

Int64 | 81.4% null | 889 unique
Links a rebound/block back to the originating shot event

reboundTotal

Int64 | 84.3% null | 32 unique
Player's running total rebound count for the game

reboundDefensiveTotal

Int64 | 84.3% null | 25 unique
Player's running defensive rebound total for the game

reboundOffensiveTotal

Int64 | 84.3% null | 17 unique
Player's running offensive rebound total for the game

blockPlayerName

String | 98.3% null | 774 unique
Name of the player who blocked the shot

blockPersonId

Int64 | 98.3% null | 941 unique
Player ID of the shot blocker

jumpBallRecoverdPersonId

Int64 | 99.7% null | 725 unique
Player ID of jump ball recoverer

stealPlayerName

String | 97.2% null | 798 unique
Name of the player who made the steal

stealPersonId

Int64 | 97.2% null | 986 unique
Player ID of the stealer

gameId

Int64 | 0.0% null | 7,281 unique
Unique game identifier (e.g. 22400001)

value

String | 99.9% null | 872 unique
Supplemental info for instant replay / review events

season

Int32 | 0.0% null | 6 unique
Season start year (e.g. 2024 = 2024-25 season)

season_type

String | 0.0% null | 2 unique
Part of season: rg = regular season, po = playoffs

rg	3,836,037	(94.3%)
po	232,269	(5.7%)

area

String | 57.1% null | 7 unique
Shot zone area from cdn.nba.com (similar to SHOT_ZONE_BASIC)

Above the Break 3	502,684	(28.8%)
Restricted Area	435,027	(24.9%)
In The Paint (Non-RA)	387,478	(22.2%)
Mid-Range	272,521	(15.6%)
Left Corner 3	76,939	(4.4%)
Right Corner 3	71,005	(4.1%)

areaDetail

String | 57.1% null | 15 unique
Detailed shot zone with distance bucket and direction

0-8 Center	660,952	(37.9%)
24+ Left Center	188,048	(10.8%)
24+ Right Center	174,900	(10.0%)
8-16 Center	137,854	(7.9%)
24+ Center	137,473	(7.9%)
8-16 Left	84,938	(4.9%)
24+ Left	78,041	(4.5%)
8-16 Right	73,991	(4.2%)
24+ Right	72,249	(4.1%)
16-24 Center	37,933	(2.2%)
16-24 Right Center	36,364	(2.1%)
16-24 Left Center	35,599	(2.0%)
16-24 Left	13,885	(0.8%)
16-24 Right	13,444	(0.8%)

isTargetScoreLastPeriod

Boolean | 36.4% null | 2 unique
Whether this event relates to a target score in the final period (always false)

False	2,588,360	(100.0%)
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shortFormattedClock

String | 99.0% null | 721 unique
Period clock in MM:SS format (only on some events)

jerseyNumber

String | 99.1% null | 55 unique
Player's jersey number (only on some events)

SHOT_ZONE_BASIC

String | 68.3% null | 8 unique
Shot zone classification from stats.nba.com (shotdetail join)

Above the Break 3	384,590	(29.8%)
Restricted Area	373,140	(29.0%)
In The Paint (Non-RA)	249,606	(19.4%)
Mid-Range	149,311	(11.6%)
Left Corner 3	67,565	(5.2%)
Right Corner 3	62,034	(4.8%)
Backcourt	2,330	(0.2%)

SHOT_ZONE_AREA

String | 68.3% null | 7 unique
Court area of the shot from stats.nba.com (shotdetail join)

Center(C)	725,416	(56.3%)
Left Side Center(LC)	169,271	(13.1%)
Right Side Center(RC)	159,271	(12.4%)
Left Side(L)	117,782	(9.1%)
Right Side(R)	114,298	(8.9%)
Back Court(BC)	2,538	(0.2%)

SHOT_ZONE_RANGE

String | 68.3% null | 6 unique
Distance range bucket from stats.nba.com (shotdetail join)

Less Than 8 ft.	520,086	(40.4%)
24+ ft.	513,981	(39.9%)
8-16 ft.	173,011	(13.4%)
16-24 ft.	78,960	(6.1%)
Back Court Shot	2,538	(0.2%)

shot_action_type

String | 68.3% null | 49 unique
Granular shot type from stats.nba.com (e.g. Driving Floating Bank Jump Shot)

Jump Shot	392,172	(30.4%)
Pullup Jump shot	155,289	(12.1%)
Driving Layup Shot	114,193	(8.9%)
Driving Floating Jump Shot	66,706	(5.2%)
Step Back Jump shot	65,136	(5.1%)
Running Layup Shot	38,415	(3.0%)
Layup Shot	33,851	(2.6%)
Driving Finger Roll Layup Shot	32,870	(2.6%)
Cutting Layup Shot	31,778	(2.5%)
Running Jump Shot	27,903	(2.2%)
Tip Layup Shot	27,332	(2.1%)
Floating Jump shot	25,126	(1.9%)
Fadeaway Jump Shot	23,732	(1.8%)
Putback Layup Shot	20,917	(1.6%)
Turnaround Fadeaway shot	19,308	(1.5%)
Driving Floating Bank Jump Shot	18,571	(1.4%)
Cutting Dunk Shot	18,537	(1.4%)
Turnaround Jump Shot	17,013	(1.3%)
Running Pull-Up Jump Shot	15,135	(1.2%)
Running Dunk Shot	15,048	(1.2%)
... and 28 more values		